

DROPZERO

Dataset Research & Data Strategy

18 Public Dataset Sources for the Retention Predictor

Purpose: Give the team and IDE/coding agent a clear list of datasets to use for DROPZERO, what each dataset is actually useful for, and what not to claim about them.

Core warning: Public video datasets usually do not contain real second-by-second YouTube audience retention. Treat them as attention, saliency, memorability, multimodal, or perceptual proxies. Real creator-provided YouTube retention should be the ground truth for final validation.

Best strategy: Use public datasets for feature learning + use creator retention data for actual retention validation.

1. Recommended Dataset List

#	Dataset	Main use	Simple explanation	Priority	Source
1	QVHighlights	Attention / important-moment proxy	Temporal highlight relevance and saliency annotations. Useful for learning which time windows are important before mapping that concept to retention risk.	High - closest proxy to temporal attention	https://github.com/jayleicn/moment_detr
2	TVSum	Human-perceived segment importance	Human ratings of important video portions. Useful for segment-level importance and attention scoring.	High - strong attention proxy	https://github.com/yalesong/tvsum
3	SumMe	Important video moments	Human-annotated important portions of videos. Useful for learning what viewers consider worth preserving.	High - attention proxy	https://zenodo.org/records/4884870
4	Memento10k	Video memorability	Memorability labels help model what types of moments are likely to stick in memory.	Medium-High - supporting signal	https://github.com/cfosco/memento_keras
5	VideoMem	Short/long-term memorability	Video memorability scores useful for a “will this moment stick?” feature.	Medium - supporting signal	https://www.interdigital.com/datasets/video-memorability-dataset
6	DHF1K	Visual attention / saliency	Eye-fixation and saliency information useful for the Visual Boredom and Visual Attention modules.	High for visual module	https://github.com/wenguanwang/DHF1K

#	Dataset	Main use	Simple explanation	Priority	Source
7	YouTube-8M	Large-scale YouTube audio/visual features	Large YouTube corpus with video/audio representations and temporal information. Useful for video representation experiments. Use a subset only.	High for representation; not retention labels	https://research.google.com/youtube8m/download.html
8	ActivityNet / ActivityNet Captions	Temporal segmentation and video-language understanding	Long videos with temporal activity annotations/captions. Useful for section boundaries and temporal understanding.	Medium-High	https://activity-net.org/download.html
9	YouCook2	Tutorial / instructional videos	Strong fit for educational content. Useful for temporal segmentation and tutorial video understanding.	High for education category	https://youcook2.eecs.umich.edu/download
10	MSR-VTT	Video-text understanding	10,000 video clips with captions. Useful for multimodal embeddings and transcript-video alignment experiments.	Medium	https://github.com/PaddlePaddle/PaddleVideo/blob/develop/docs/en/dataset/msrvtt.md
11	MSVD / MSVD-QA	Video-language understanding	Video descriptions and QA useful for multimodal semantic understanding and reasoning.	Medium	https://github.com/salesforce/LAVIS/blob/main/dataset_card/msvd_qa.md
12	How2	Instructional speech + video	Instructional videos with subtitles/summaries. Useful conceptually for speech, transcript, and tutorial content modeling; video access may require additional handling.	Medium-High	https://github.com/srvk/how2-dataset

#	Dataset	Main use	Simple explanation	Priority	Source
1 3	TGIF-QA	Temporal reasoning	Question-answering over GIF/video sequences. Useful for temporal understanding and event reasoning.	Medium	https://github.com/YunseokJANG/tgif-qa
1 4	QVHighlights Videos	Raw videos for QVHighlights	Useful when you want to run your own frame/audio extraction against QVHighlights annotations.	High when using QVHighlights	https://huggingface.co/datasets/ayushsdev/qvhighlights-videos
1 5	LIRIS-ACCEDE	Emotion / arousal / valence	Affective annotations can support audio-visual engagement-related features. Do not use as direct retention labels.	Medium - supporting feature	https://liris-accede.ec-lyon.fr/
1 6	KoNViD-1k	User-generated video quality	Human subjective video-quality scores. Useful as a supporting signal so poor visual quality is not confused with content-driven retention risk.	Medium - video quality	https://database.mmsp-kn.de/konvid-1k-database.html
1 7	LIVE Video Quality Database	Perceived video quality	Human subjective quality ratings. Useful for quality-aware analysis.	Medium - supporting signal	https://live.ece.utexas.edu/research/Quality/live_video.html
1 8	LSVQ	Real-world/social video quality	Large-scale UGC video quality dataset for perceptual quality features.	Medium - supporting signal	https://github.com/baidut/PatchVQ

2. What Each Dataset Should Do Inside DROPZERO

Module	Datasets	Use inside product
Retention / Attention model	QVHighlights + TVSum + SumMe	Train/test segment-importance or attention proxy models. These are not direct YouTube retention labels.
Visual attention	DHF1K	Support visual saliency features, visual boredom detection, and visual-change reasoning.
Memorability	Memento10k + VideoMem	Support a memorability signal: moments that may remain memorable can be useful as an auxiliary engagement feature.
YouTube video representation	YouTube-8M	Useful for large-scale audio/video representation experiments. Do not use its topic labels as retention ground truth.
Tutorial / education	YouCook2 + How2	Useful when testing educational/tutorial video analysis, segmentation, and transcript alignment.
Video-text understanding	MSR-VTT + MSVD-QA + QVHighlights	Useful for semantic alignment between transcript, video, questions, actions, and visual content.
Emotion / arousal	LIRIS-ACCEDE	Optional support signal for affective/audio-visual analysis.
Video quality	KoNViD-1k + LIVE + LSVQ	Use as control variables so poor video quality does not get mistaken for content-driven retention problems.
Actual retention validation	Creator-provided YouTube retention	This is the most important ground-truth source. Compare actual retention with DROPZERO prediction and measure error/detection performance.

3. The Correct Data Architecture

Do not merge all datasets blindly. They represent different concepts.

Public proxy data

QVHighlights / TVSum / SumMe → attention importance

DHF1K → visual attention

Memento10k / VideoMem → memorability

YouTube-8M → audio/video representations

YouCook2 / How2 → tutorial video understanding

LIRIS-ACCEDE → affective signals

KoNViD-1k / LIVE / LSVQ → perceived video quality

Ground truth

Creator video + actual YouTube retention → real retention curve used for validation.

4. Recommended Hackathon Download Order

#	Dataset	Why now
1	QVHighlights	Start here because it is closest to temporal attention/highlight relevance.
2	TVSum	Add human segment-importance annotations.
3	YouCook2	Useful if your demo includes tutorial/educational videos.
4	Memento10k	Add memorability as a supporting signal.
5	DHF1K	Build the visual attention/visual boredom module.
6	YouTube-8M subset	Only download a manageable subset for audio/video representations.
7	Creator retention dataset	Collect the team's real validation data and use it as the true retention target.

Do not download everything. Large datasets can consume huge amounts of time and storage. For a 24-hour hackathon, prioritize small/medium annotation datasets and use subsets of very large datasets.

5. Suggested Training / Validation Setup

Stage A - Public proxy learning: learn segment importance, saliency, memorability, video-text alignment, and other supporting signals.

Stage B - Feature extraction on creator videos: for every 5-15 second segment, compute text, audio, visual, pacing, structure, repetition, novelty, and promise/payoff features.

Stage C - Real retention validation: align actual retention timestamps with your segments and train/calibrate the retention-risk model on creator data.

Stage D - Holdout evaluation: keep entire videos out of training when evaluating. Do not randomly split segments from the same video between train and test, because that can leak video-specific patterns.

6. Suggested Folder Structure

```
data/
  raw/
    qvhighlights/
    tvsum/
    summe/
    dhf1k/
    memento10k/
    youtube8m_subset/
    creator_retention/
  processed/
    segments.parquet
    features.parquet
    retention_aligned.parquet
  models/
    retention_model.pkl
    scaler.pkl
    feature_schema.json
```

7. Important Claims the Team Can Safely Make

1. “We use public datasets to learn proxies for attention, visual saliency, memorability, and video understanding.”
2. “Our true retention validation uses creator-provided audience-retention data.”
3. “We compare the predicted and actual curves at the same time resolution.”
4. “We detect likely drop-off zones using a combination of learned model output and measurable evidence.”
5. “Our before/after numbers are model simulations, not guaranteed audience outcomes.”
6. “As more creator retention data is collected, the system can be calibrated and personalized.”

8. Final Recommendation

For the hackathon, focus on QVHighlights + TVSum + DHF1K + Memento10k + a small YouTube-8M subset, then build your strongest proof around your own creator-retention validation set. The public datasets make the system multimodal and research-backed; the creator data makes the retention claim testable.

Dataset use should also respect each dataset’s license, access conditions, and intended research/commercial use before inclusion in a public or commercial deployment.